

Group 6

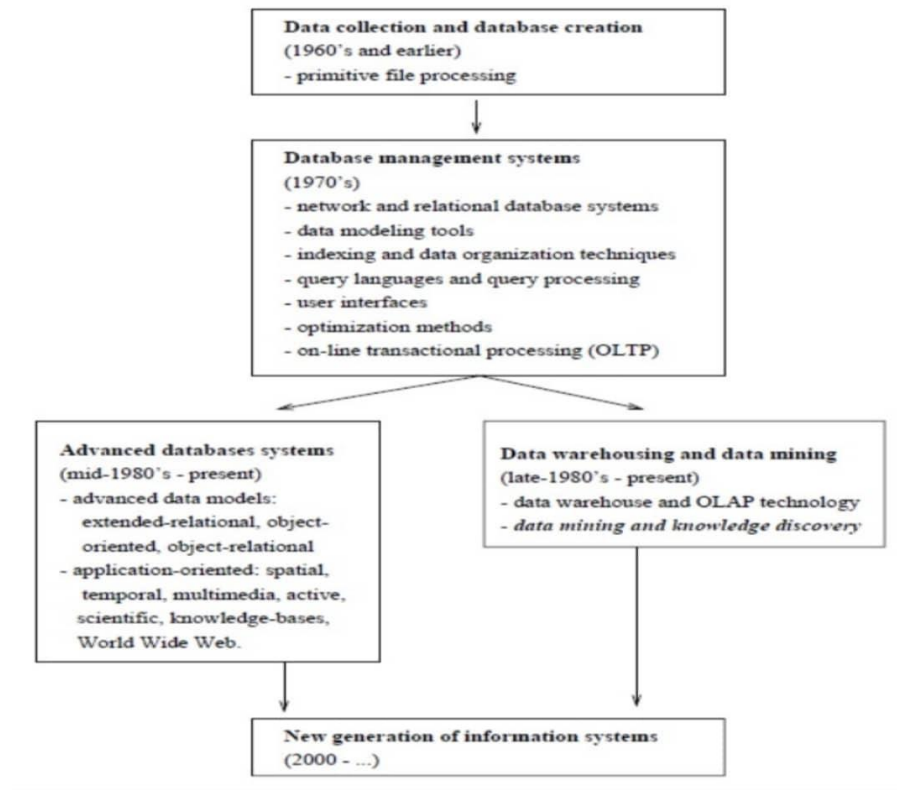
1. CHERYL CHEONG KAH VOON
2. WOO CHENG SHUAN
3. BRENDAN CHIA YAN FEI

Group Task 1

Q1 - How is data mining part of the nature evolution of database technology?

Data Mining can be viewed as result of the nature evolution of information technology (the database industry).

- In 1960's and earlier
Data collection and database creation using primitive file processing
- 1970s
Data management system introduced network and relational database system, data modeling tools, indexing and data organization techniques, query language and query processing, user interface, optimization method and on-line transactional processing
- Mid-1980s to present
Advanced database systems emerged, featuring extended-relational, object-oriented models and application-specific databases like multimedia and spatial databases
- Late 1980s to present
Data warehousing and data mining became prominent, focusing on OLAP technologies and knowledge discovery from large datasets.
- 2000 till now
A new generation of information systems developed



Q2- Why is data mining important?

Data mining is important because it enables the discovery of meaningful patterns and relationships in large volumes of datasets in a relatively quick and efficient way. Key reasons for its importance include pattern recognition, data-driven decision-making, customer insights and various other applications.

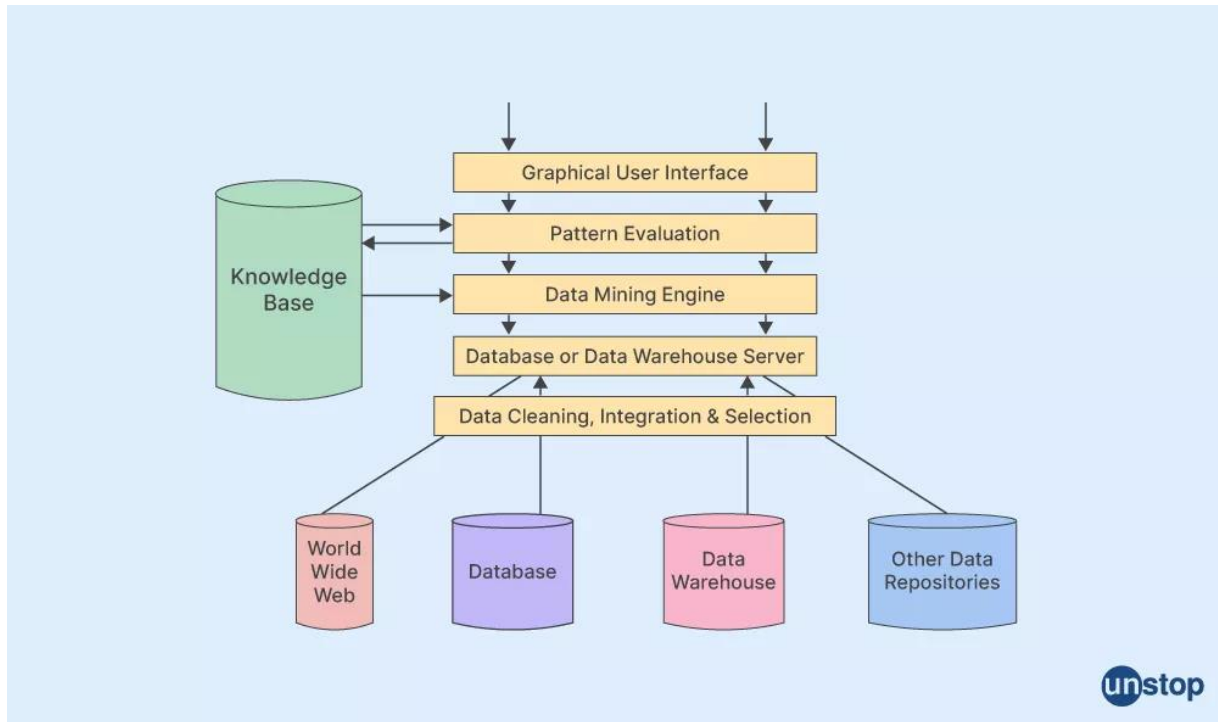
Q3- What is data mining? (Provide a clear definition)

Data mining is the use of machine learning and statistical analysis to uncover patterns and other valuable information from large data sets.

[Source](#)

Q4- What is the general architecture of a data mining systems?

This layered architecture shows how a data mining system takes in data from diverse sources, processes and prepares it, mines useful patterns using algorithms, evaluates the results, presents the findings to users through an interface and stores learned knowledge for future use.



[Source](#)

Q5- What types of data can be mined? (Structured & Unstructured data)

Structured data	Unstructured data
Stored in tabular form (spreadsheets) and managed in relational database	Includes various content such as documents, videos, audio files and emails
Pros: -easy to use for business users -extensive tools support -instantly usable	Pros: -native format -flexible -lower cost
Cons: -data preparation -not flexible -high overhead -complex data structures	Cons: -lack of visibility -requires advanced analytics -requires dedicated tools

Q6- What are the main steps/processes involved in data mining?

The main steps involved in data mining are data selection, data process, data transformation, data mining and evaluation.

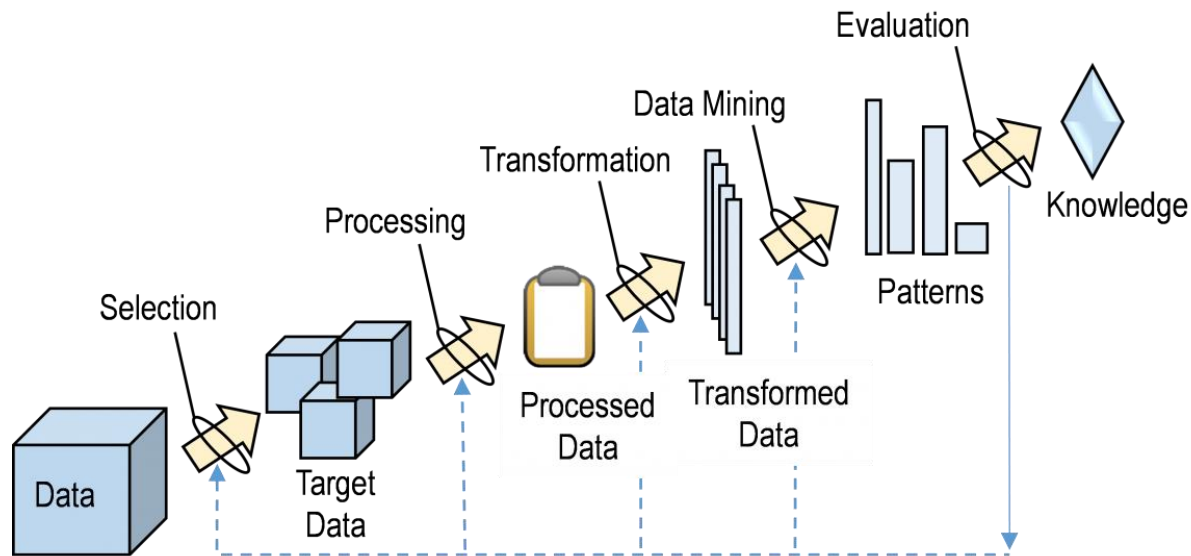


Figure 6: Main process in data mining

Q7- What are the common data mining tasks/techniques and their associated algorithms?

- **Association rule mining** - if-then statements that identify relationships between data elements
- **Classification** - Decision trees, Naive Bayes classifiers, k-nearest neighbors (KNN) and logistic regression
- **Clustering** - k-means clustering, hierarchical clustering and Gaussian mixture models
- **Regression** - Linear regression and multivariate regression
- **Sequential Pattern** - look for patterns in which a particular set of events or values leads to later ones

[Source](#)

Q8- What Issues or challenges are commonly encountered in data mining?

- **Data quality** – noisy data, incomplete data and unstructured data can significantly impair the effectiveness of data mining measure
- **Data privacy and security** – unauthorized access to data can lead to privacy and data security breaches and misuse of personal information
- **Handling complex and diverse data types** – often challenged by the need to process complex data types such as spatial data, temporal data and media data
- **Scalability and efficiency** - with the growth of data volume, scalability and efficiency refining data mining requests become critical issues.

Group Task 2

Q1 – What is Knowledge Discovery in Databases (KDD)?

KDD is a systematic approach to uncovering patterns, relationships and actionable insights from vast datasets. It involves multiple steps, from selecting and preprocessing data to the actual process of data mining and finally to the interpretation and use of the results.

Q2 – What are the steps in the KDD process? (Include a figure/diagram)

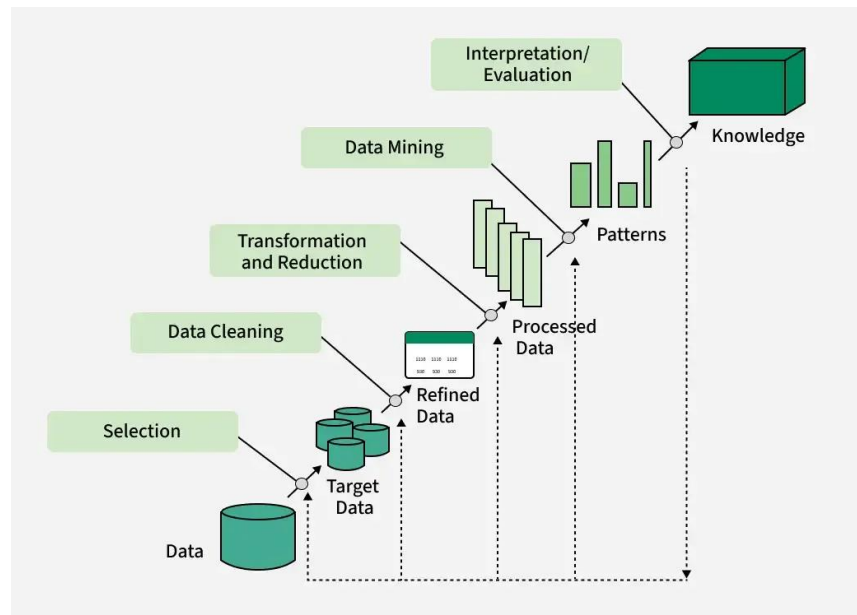


Figure 2 : KDD Process

1. Data Selection
2. Data Cleaning and Preprocessing
3. Data Transformation and Reduction
4. Data Mining
5. Evaluation and Interpretation of Results

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Q3 – How does KDD differ from Data Mining? (KDD vs. Data Mining)

KDD vs Data Mining

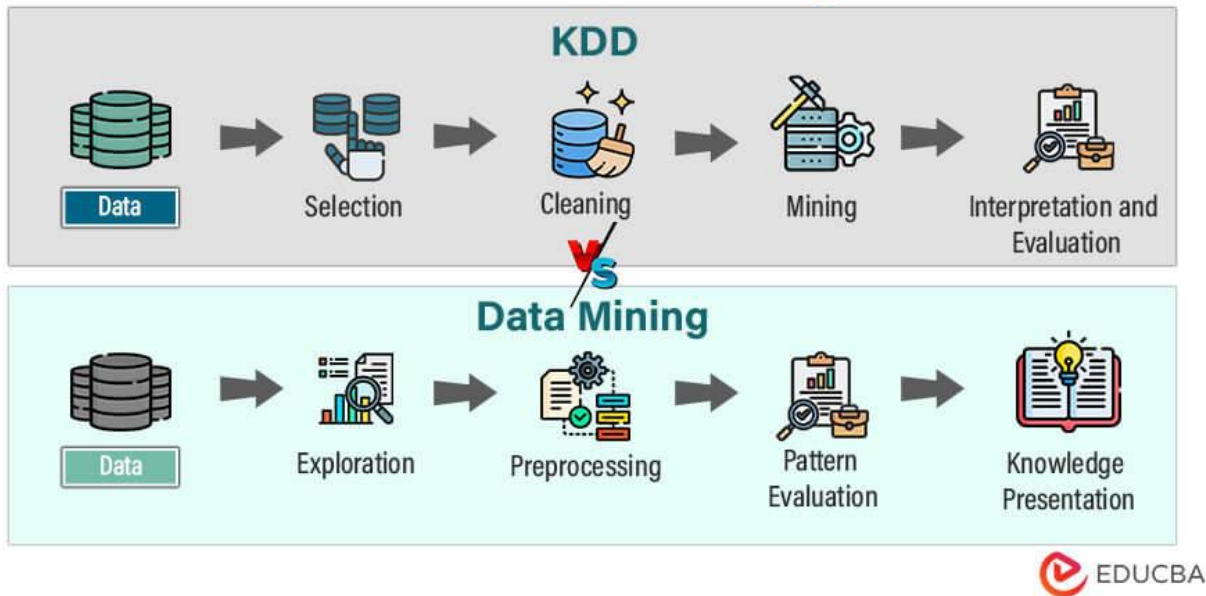


Figure 6: KDD vs Data Mining

Section	KDD	Data Mining
Scope	The broader process includes data mining as a step	A specific process within KDD focused on data analysis
Definition	Comprehensive process of knowledge extraction	A specific technique for discover patterns in data
Stages	Includes data selection, preprocessing, transformation, data mining, interpretation and knowledge presentation	Primary involves data mining techniques
Application	Used for comprehensive knowledge extraction in various domains	Applied to specific tasks such as classification, clustering and association rule mining
Example usage	Business intelligence, healthcare analytics, fraud detection	Customer segmentation, predict modeling, anomaly detection

Q4 – What is Data and Insights?

-Data refers to raw, unprocessed facts and figures without context. These can be in the form of numbers, text, images, or other formats. For example, a list of temperatures recorded each day constitutes data.

-Insights are the valuable understandings or conclusions drawn from analyzing data. They go beyond mere presentation of information to reveal underlying patterns, trends, or relationships that can inform decision-making. For instance, analyzing temperature data over a month to identify climate trends provides insights into weather patterns.

Q5 –What is the difference between data and insights?

Aspect	Data	Insights
Definition	Raw, unprocessed facts (e.g., numbers, text, images)	Deeper understanding of patterns, relationships, and actionable opportunities derived from analyzed information
Purpose	Provides raw material for analysis	Guides decision-making by explaining "why" trends occur and suggesting solutions
Example	Website traffic counts, customer purchase histories	"Customer satisfaction improved after implementing AI chatbots"

Q6 – What is CRISP-DM (Cross Industry Standard Process for Data Mining)?

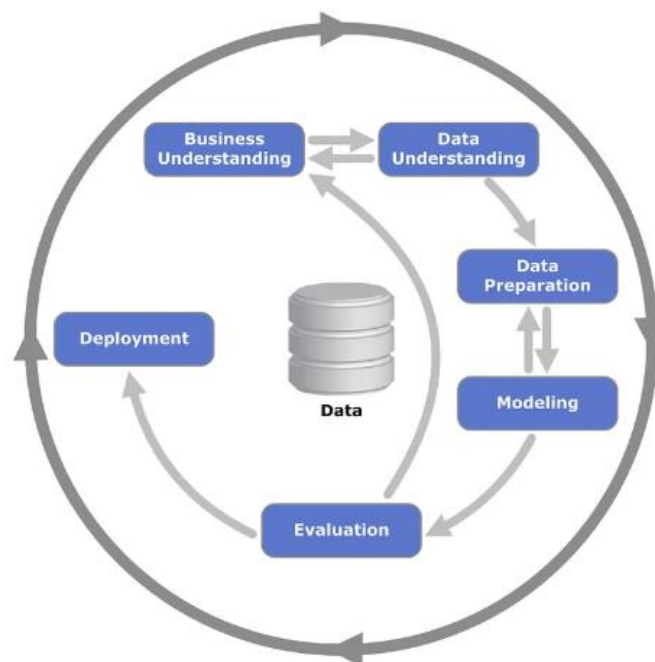


Figure 6 : Diagram of CRISP-DM Process

CRISP-DM is a cyclical process that provides a structured approach to planning, organizing, and implementing a data mining project. The process consists of six major phases as shown in the diagram above

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Q7 – How does Information Retrieval (IR) differ from Data Mining?

Information Retrieval (IR)	Data Mining
works with unstructured or semi-structured data	deals with structured or numerical data
mainly relies on statistical, linguistic, or heuristic methods	focuses on machine learning, artificial intelligence, or mathematical methods
focus on relevance, usefulness, or satisfaction	concentrates on accuracy, performance, or generalization

[Source](#)

Q8 – What is Machine Learning (ML)?

A: Machine learning is a subset of artificial intelligence (AI). With machine learning, computers analyse large data sets and then ‘learn’ patterns that will help it make predictions about new data sets. Apart from the initial programming and maybe some fine-tuning, the computer doesn’t need human interaction to learn from the data.

Q9 – What are the main types of ML?

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Q10 – How is DM different from ML (DM vs. ML)?

Data Mining	Machine Learning
Discovery of hidden patterns or knowledge from data	Development of algorithms that learn from data
Extract insights and information from existing datasets	Build models to make predictions or perform tasks
Identifying patterns, trends, and anomalies	Predictive modeling, classification, clustering

[Source](#)

Q11 – What is Artificial Intelligence (AI)?

A : AI is a technology that enables computers and machines to simulate (to imitate or replicate a process) human learning, comprehension, problem solving, decision making, creativity and autonomy.

[Source](#)

Q12 – What is the difference between ML and AI (ML vs. AI)?

AI	Machine Learning
allows a machine to simulate human intelligence to solve problems	allows a machine to learn autonomously from past data
develop an intelligent system that can perform complex tasks	build machines that can learn from data to increase the accuracy of the output
technologies in a system so that it mimics human decision-making	self-learning algorithms to produce predictive models

[Source](#)

Q13 – What are some real-world applications of AI and potential careers in this field?

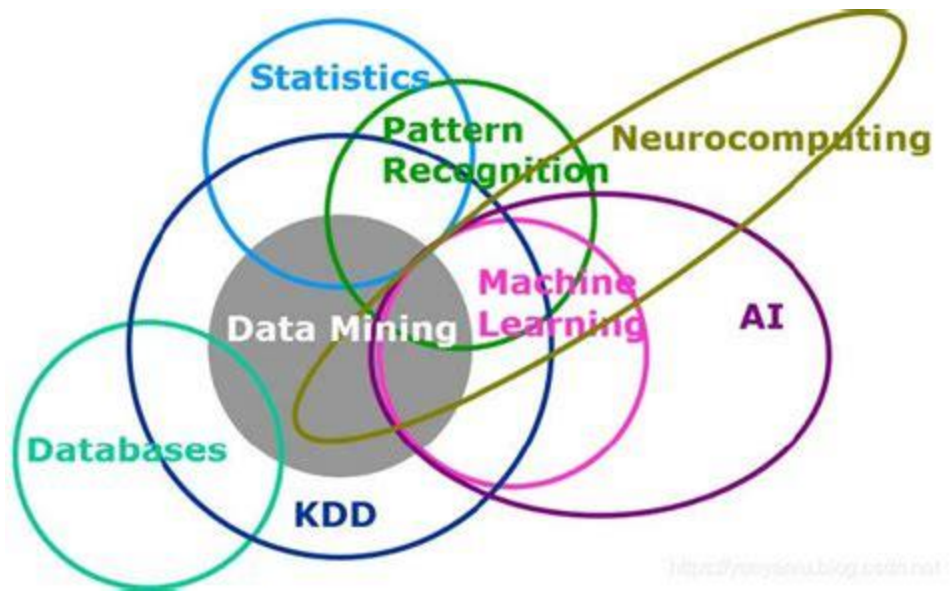
Real-World Application

- Face Recognition
- E-Commerce
- AI in Healthcare
- Agriculture

Potential Careers

- AI Internships
- Junior Data Analyst
- Machine Learning Intern

Q14 – Provide a figure that illustrates the relationships among KDD, DM, ML and AI.



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